

LE HAVRE-OCTEVILLE AP, France

WMO: 070460

Lat: 49.5331N

Lon: 0.0917E

Elev: 308

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.3	29.0	19.4	14.8	29.8	22.6	17.2	30.7	50.7	46.4	42.6	48.6	9.0	70	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	81.0	66.7	76.5	64.6	72.9	63.0	68.3	77.9	66.3	73.3	64.7	70.5	10.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.6	71.3	63.6	88.9	69.2	62.2	84.8	67.3	32.7	78.2	31.1	73.4	29.8	70.7	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.3	25.9	22.4	DB	23.2	89.5	3.9	5.5	20.4	93.4	18.1	96.6	16.0	99.7	13.1	103.6	(4)
(5)			WB	21.9	71.5	4.1	2.1	18.9	73.0	16.5	74.2	14.3	75.4	11.3	76.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	42.0	41.9	45.8	50.2	55.0	60.6	63.9	63.9	60.6	55.3	48.4	43.5		
	DBStd	9.59	6.31	6.08	5.52	5.57	5.23	4.90	4.24	3.79	4.15	5.30	5.13	6.13		
	HDD50	1019	252	230	152	63	9	0	0	0	0	16	86	211		
	HDD65	4615	714	646	594	446	314	153	72	65	143	303	497	668		
	CDD50	1983	3	4	22	67	164	317	429	432	318	179	39	9		
	CDD65	106	0	0	0	0	5	19	37	32	11	2	0	0		
	CDH74	683	0	0	0	5	34	165	247	171	54	7	0	0		
	CDH80	180	0	0	0	0	3	45	73	48	10	1	0	0		
(14) Wind	WSAvg	11.2	13.1	12.6	11.8	10.3	10.3	9.8	9.9	9.8	9.9	11.0	12.6	13.1		
(15) Precipitation	PrecAvg	33.5	2.9	2.4	2.1	2.2	2.3	2.5	2.2	2.7	2.8	3.7	3.8	3.9		
	PrecMax	44.3	6.2	4.7	7.1	4.6	4.2	5.6	4.9	6.9	6.3	8.1	8.7	8.1		
	PrecMin	20.9	0.3	0.5	0.4	0.2	0.5	0.4	0.4	0.4	0.6	0.4	0.9	0.9		
	PrecStd	4.9	1.4	1.3	1.4	1.3	1.0	1.4	1.1	1.7	1.3	1.9	1.8	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	58.5	65.8	73.6	78.3	85.8	87.8	86.2	80.6	72.7	62.2	56.7		
		MCWB	52.1	49.5	53.9	58.0	63.8	68.7	69.6	68.8	65.0	62.5	58.5	53.1		
	2%	DB	53.2	54.0	60.8	67.6	72.7	78.8	80.3	78.7	74.0	67.7	59.0	54.7		
		MCWB	51.0	49.3	52.1	56.5	61.0	66.2	67.2	66.1	63.0	61.4	55.5	52.0		
	5%	DB	51.3	51.8	56.7	62.8	67.9	73.3	75.3	73.6	70.2	64.3	57.1	53.1		
		MCWB	49.3	48.1	50.5	54.0	58.6	63.2	64.5	64.2	61.7	59.7	54.1	50.9		
	10%	DB	49.7	49.8	53.5	58.8	63.8	69.0	71.3	70.1	67.2	62.2	55.3	51.4		
		MCWB	47.8	46.8	49.2	52.0	57.1	61.9	62.9	63.0	61.1	58.4	52.7	49.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	52.0	55.7	60.4	66.4	70.3	70.8	70.6	68.2	64.0	59.8	54.2		
		MCDB	53.9	54.7	62.9	69.8	75.5	82.3	83.8	82.4	75.0	68.6	61.5	55.5		
	2%	WB	51.5	50.4	53.3	56.8	62.3	67.0	68.2	67.5	65.2	62.1	56.3	52.7		
		MCDB	52.7	53.0	58.0	65.4	69.3	76.3	78.2	75.2	70.2	66.4	58.3	54.2		
	5%	WB	49.8	48.8	51.5	54.6	59.7	64.6	65.9	65.5	63.6	60.3	54.6	51.3		
		MCDB	51.2	50.9	55.6	61.9	66.0	72.1	72.5	71.2	68.0	63.5	56.5	53.0		
	10%	WB	48.1	47.3	49.6	52.6	57.6	62.5	64.1	63.9	62.2	59.0	53.1	49.5		
		MCDB	49.5	49.4	53.0	57.9	63.2	67.6	69.8	68.5	66.1	61.8	55.0	51.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.3	10.0	12.2	11.6	12.4	11.5	10.5	11.4	9.4	7.4	7.2		
		MCDBR	6.9	11.1	15.5	19.2	19.4	21.2	20.8	18.6	17.3	11.7	8.1	7.1		
	5% WB	MCWBR	6.7	7.9	9.4	10.8	10.8	11.2	10.2	9.2	8.9	7.3	6.6	6.5		
		MCWBR	6.6	8.6	13.3	17.2	16.9	18.9	17.6	14.7	13.1	10.6	7.7	6.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.327	0.344	0.378	0.396	0.401	0.400	0.395	0.390	0.374	0.359	0.334	0.321		
		taud	2.431	2.408	2.313	2.283	2.313	2.346	2.363	2.398	2.426	2.456	2.451	2.400		
	Ebn at Noon	Ebn at Noon	228	253	262	269	272	273	272	269	261	244	223	212		
		Edh at Noon	20	26	34	38	39	38	37	34	30	25	20	18		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	280	513	883	1381	1624	1812	1737	1460	1122	642	338	233		
	RadStd	RadStd	27	66	100	138	132	175	121	108	70	56	36	30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page